

nlmixr² Current Status

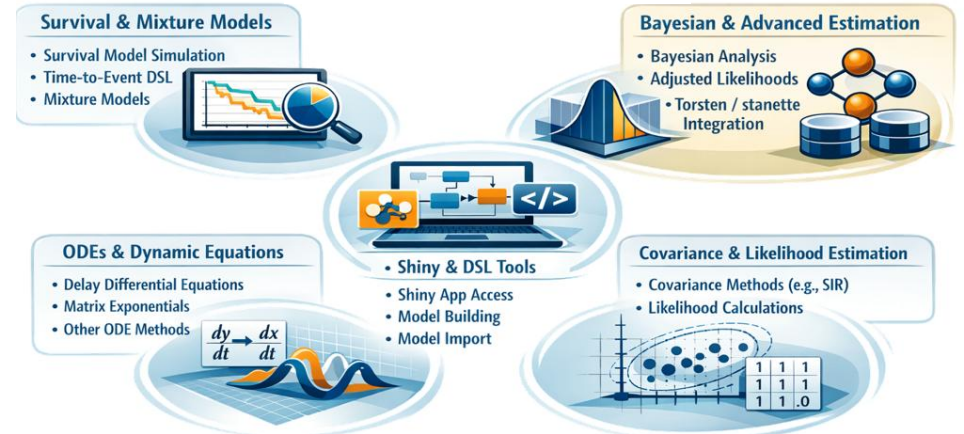
Outline



nlmixr² Integration



nlmixr² Future



nlmixr² : estimation methods



Naïve pooled estimation enables rapid model development

Common Model Structure

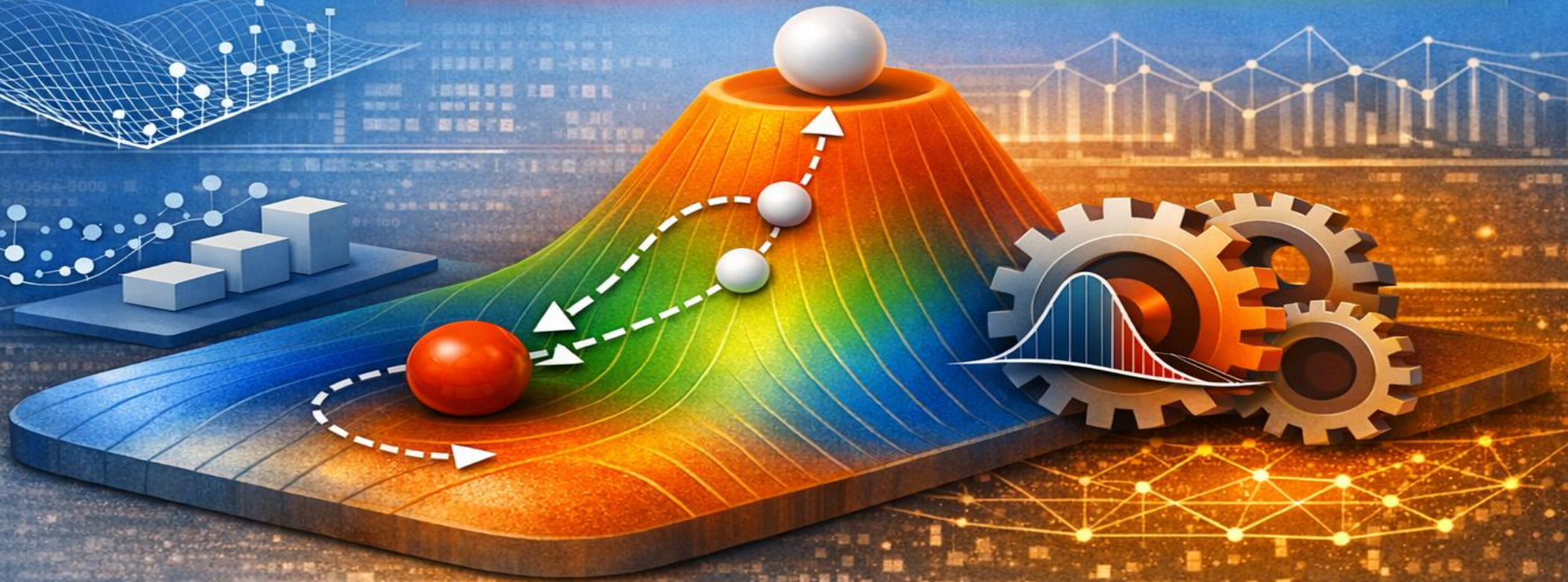
- Same model specification and fit object as mixed-effects

Fast Optimization

- Derivative-free and gradient-based methods

Seamless Transition

- Models extend directly to mixed-effects estimation



Optimization methods supporting naïve pooled workflows

Deterministic and likelihood-based optimizers supporting baseline and diagnostic workflows

- **Derivative-free optimization methods**
 - bobyqa, uobyqa (from minqa)
 - Nelder-Mead, SANN, Brent (base R via optim)
- **Gradient-based methods**
 - nls, nlminb, nlm (from base R)
 - L-BFGS-B, BFGS, CG (base R via optim)
 - lbfgsb3c, n1qn1, trust (imported in nlmixr2)
 - fmeMcmc, pseudoOptim (used with fme/babelmixr2)
 - Note the “focei” method also supports naïve pooled estimation, changing to a likelihood optimization
- **All Naïve pooled estimation methods except nls/focei support generalized likelihoods**

Transitioning from naïve pooled to Nonlinear Mixed-Effects Modeling

Common Model Structure

- Same structural model, error model, and parameterization
- Same model code and data structure used throughout development
- Naïve pooled and NLME share consistent likelihood framework

Continuity in Estimation Framework

- Smooth transition in estimation approach
- Provides a natural stepping stone to population inference
- Reduces re-coding and model reformulation between stages

Nonlinear Mixed-Effects Estimation

- Core NLME methods supported
- FOCEI — gradient-based likelihood estimation
- Adaptive Gaussiav Quadrature
- SAEM — stochastic approximation



Designed to support model development workflows that scale naturally from exploratory analyses to population modeling

Nonlinear Mixed-Effects Estimation Methods (Implemented)

Supported Estimation Methodologies

- ◆ **FOCEI**
 - ◆ Likelihood-based estimation
 - ◆ Gradients and bounded optimization
- ◆ **Adaptive Gaussian Quadrature (AGQ)**
 - ◆ Likelihood-based estimation *for nonlinear mixed-effects models*
- ◆ **Laplace Approximation**
 - ◆ Likelihood-based *alternative to AGQ*
- ◆ **SAEM**
 - ◆ Stochastic Approximation EM
 - ◆ Parallel execution
 - ◆ Mu-referencing *formulation*



Methods Under Active Development or Evaluation

- ◆ **adFOCEI** – Automatic differentiation for inner gradients
- ◆ **muFOCEI** – Linear mu-referencing for population effects
- ◆ **Importance Sampling (IMP / IMPMAP)**
- ◆ **f-SAEM** – Stochastic approximation extensions

More technical details about babelmixr2 integrations



Method	Import	Notes
NONMEM	nonmem2rx	Creates NONMEM control stream, runs and imports into nlmixr2 • Can help diagnose NONMEM force issues • Validates translation (also in the import) • Export with the "nonmem" estimation method
Monolix	monolix2rx	Creates Monolix project, runs and imports into nlmixr2 • Can add CWRES to Monolix model • Validates translation (also in the import, work in progress) • Export with the "monolix" estimation method
PopED	--	Creates a PopED database, • combining the ease of using a nlmixr2 model and a simple dataset structure for optimal design
fmeMcmc	FME	• A full estimation method that relies on the external package fme • Shows how nlmixr2 can be extended to estimation methods outside its core functionality.
pseudoOptim	FME	
saemix	saemix	• Uses a nlmixr2 function to run saemix (as supported by saemix)
nlmer	lme4	• Uses a nlmixr2 function to support gradient creation and running with nlmer

nlmixr² roadmap

Survival & Mixture Models

- Survival Model Simulation
- Time-to-Event DSL
- Mixture Models



Bayesian & Advanced Estimation

- Bayesian Analysis
- Adjusted Likelihoods
- Torsten / stanette Integration



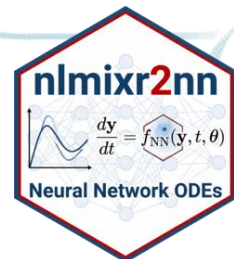
nlmixr²bayes

Powered with Stan 

ODEs & Dynamic Equations

- Delay Differential Equations
- Matrix Exponentials
- Other ODE Methods

$$\frac{dy}{dt} \rightarrow \frac{dx}{dt}$$



Shiny & DSL Tools

- Shiny App Access
- Model Building
- Model Import



Covariance & Likelihood Estimation

- Covariance Methods (e.g., SIR)
- Likelihood Calculations



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1	1	1
1	1	.0

nlmixr² acknowledgements

- Nlmixr team past & present
- Advisory Committee
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 - Irina Baltcheva
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 - Mick Looby
 - Kai Grosch
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- External Advocates



Thank you

